

BYOD: BUILD YOUR OWN DLM - EFFICIENT AUTOREGRESSIVE-TO-DIFFUSION LANGUAGE MODEL CONVERSION

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ABSTRACT

Diffusion language models (DLMs) permit parallel token updates and flexible generation orders, but leading systems rely on costly pretraining or continued full-model training. We ask how efficiently diffusion generation can instead be induced in existing instruction-tuned autoregressive models. Build Your Own DLM (BYOD) replaces causal attention with bidirectional attention and trains only query/value LoRA adapters under a masked denoising objective, leaving 94.2–95.9% of parameters frozen. Without architecture-specific tuning, the same recipe converts Gemma 2 9B, Llama 3.1 8B, Qwen2.5 7B, and Ministral 8B, with each 25,000-update conversion processing at most 102.4M token positions in 3.1–4.0 hours on one Google Colab GPU. All four models generate from fully masked sequences, use right context in a controlled attention test, and retain substantial parent-model knowledge. BYOD-Gemma is the strongest conversion on our nine-task evaluation, reaching 53.0% average accuracy compared with 51.9% for LLaDA 8B Instruct in the same harness. Open-generation quality remains the principal limitation: at 64 denoising network evaluations (NFE), converted-model perplexity is 8.3–11.1 versus 4.1 for LLaDA and 2.2–3.0 for the AR parents, with generally lower Distinct- n scores as well. BYOD therefore establishes a practical, reproducible route for broad experimentation with diffusion generation while identifying the quality gap that efficient conversion must next close.

1 INTRODUCTION

Autoregressive (AR) language models generate a sequence from left to right. This method has proven extremely effective due to its simple learning target and effective scaling capabilities. However, it also has trade-offs: it fixes the generation order, limits the model to unidirectional attention, and predicts only one new token at a time. Diffusion language models (DLMs) instead begin with a corrupted sequence and can iteratively reconstruct multiple positions at a time using bidirectional context (Austin et al., 2021a; Sahoo et al., 2024; Shi et al., 2024). Their number of function evaluations (NFE) can be chosen independently of output length, and early low-probability predictions can be revised. However, training or adapting models for diffusive generation has so far been computationally intensive and therefore expensive.

Our primary aim is therefore to devise a recipe optimized foremost for compute efficiency and reproducibility when converting AR models into DLMs, rather than to pretrain a new diffusion foundation model for every architecture. Efficiency matters for three connected reasons. First, full diffusion pretraining is financially and environmentally costly: under our standardized rental and electricity scenario, the reported LLaDA run corresponds to approximately \$675,000 and 44,100 kg CO₂e (Table 1). These are not the authors’ bill or measured emissions—operational carbon depends strongly on utilization, datacenter efficiency, location, and energy supply—but they illustrate the scale of the barrier (Strubell et al., 2019; Patterson et al., 2021). Second, a conversion that completes in hours on one widely available cloud instance makes hands-on DLM research accessible beyond laboratories with multi-node clusters. Third, applying the same inexpensive intervention to multiple existing model families makes it feasible to study how architecture and parent-model choice affect diffusion generation without repeating foundation-model pretraining for every comparison.

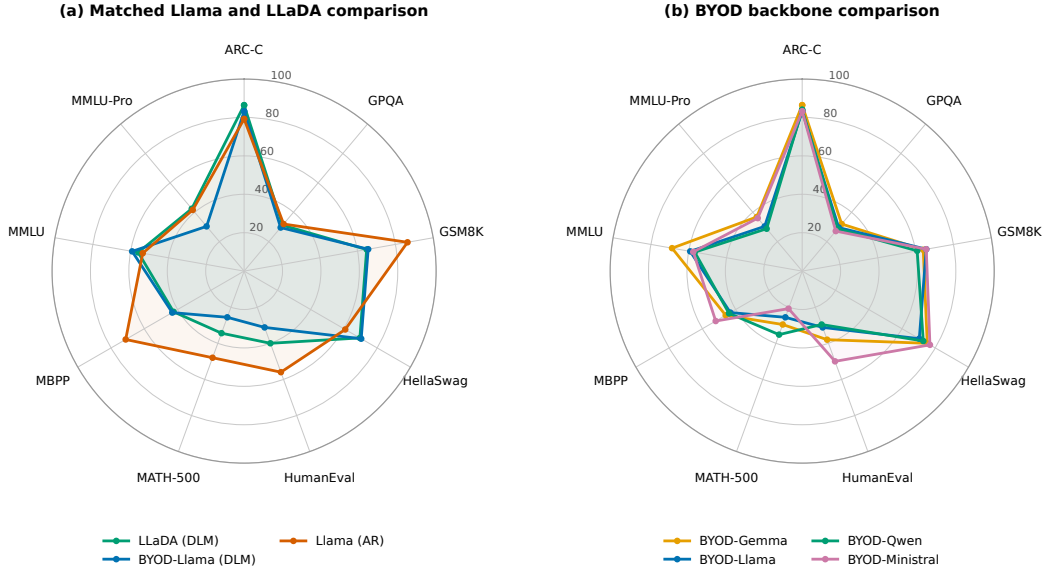


Figure 1: Task-level accuracy profiles from the matched evaluation. (a) LLaDA, BYOD-Llama, and the unchanged Llama AR parent. (b) The four primary BYOD conversions.

We study this question with BYOD, a generalizable conversion and evaluation pipeline for four instruction-tuned decoder families: Gemma 2 9B, Llama 3.1 8B, Qwen2.5 7B, and Ministral 8B. We selected the latest text-only instruction checkpoint from each family with a comparable parameter count. Multimodal variants are not used to avoid modality-specific training as an additional confounder. Every backbone receives the same bidirectional attention intervention, LoRA target modules, and masked-diffusion objective. This paired design separates conversion quality from the strength of the starting checkpoint. We directly compare our models against LLaDA 8B Instruct because its standard masked-diffusion objective, instruction-tuning setup, and overlapping benchmark suite make it the cleanest external control. We additionally report published DiffuLLaMA, Dream-Instruct, iLLaDA-Instruct, and Efficient-DLM values, but keep them protocol-separated: their additional attention, masking, scheduling, data, and inference techniques would confound a controlled comparison.

Our contributions are:

- a compute-efficient recipe that converts four instruction-tuned AR model families to masked diffusion using only a few single-GPU hours and LoRA parameter updates;
- a paired evaluation of each diffusion model and its frozen AR parent across general knowledge, reasoning, mathematics, and code benchmarks;
- a controlled analysis of training duration, denoising-step count, validation loss, perplexity and distinct n-grams; and
- publicly available training and inference code directly available through Google Colab and Github, and interactive demos for all four models, making DLM conversion and qualitative evaluation easily accessible.

2 RELATED WORK

Discrete diffusion models corrupt and reconstruct categorical sequences rather than adding Gaussian noise. D3PM formalized categorical corruption with transition matrices, including an absorbing process in which tokens transition to a dedicated mask state (Austin et al., 2021a). MDLM and contemporaneous work showed that this masked formulation admits a simple noise-weighted cross-entropy objective, while RADD reparameterized the predictor in terms of conditional distributions over clean tokens (Sahoo et al., 2024; Shi et al., 2024; Ou et al., 2025). These approaches form the

Table 1: Incremental training scale and standardized resource estimates for comparable diffusion systems. GPU-hours marked [†] are estimated from published model/data scale; unmarked values are reported. We stress that cost and CO₂e are estimates purely for illustrative purposes, based on a standardized operational scenario (Appendix A), not reported values. NR: insufficient information.

System	Diffusion-stage data	Updated weights	Hardware	GPU-h	Cost (USD)	CO ₂ e
BYOD (ours)	$\leq 102.4\text{M}$ tokens	LoRA	1 $\times$ RTX Pro 6000	3.2	\$11	0.9 kg
DiffuLLaMA 7B	65B tokens	Full	64 $\times$ GH200	1,917 [†]	\$9,951	0.65 t
LLaDA 8B	2.3T tokens	Full	H800	130,000	\$674,830	44.1 t
Dream 7B	580B tokens + SFT	Full	NR	17,105 [†]	\$88,791	5.81 t
iLLaDA 8B	12T tokens + SFT	Full	NR	404,449 [†]	\$2,099,494	137.3 t
Efficient-DLM 8B	500B tokens	Full	128 $\times$ H100	16,852 [†]	\$87,479	5.72 t
Uno 8B	7B tokens	LoRA	64 $\times$ H200	3,840	\$19,933	1.30 t
Fast-dLLM v2 7B	$\sim 1\text{B}$ tokens	Full	64 $\times$ A100	768	\$3,987	149 kg

basis of masked diffusion language models (MDLMs). Uniform (or multinomial) diffusion, which instead corrupts tokens with samples from the ordinary vocabulary, is another option (Hooeboom et al., 2021); we use the simpler absorbing-mask denoising objective employed by LLaDA.

The authors of LLaDA were the first to show that masked diffusion can scale to an 8B-parameter general-purpose language model (Nie et al., 2025). Although its architecture follows Llama 3.1 8B, LLaDA was pre-trained from scratch using 2.3T tokens and 130,000 GPU-hours. AR-to-diffusion conversion instead amortizes knowledge already acquired by an AR checkpoint. DiffuLLaMA established this route through 65B tokens of full-model continued training at 7B scale (Gong et al., 2025). Dream adopts DiffuLLaMA’s shifted-prediction initialization, continues Qwen2.5 for 580B tokens, and adds context-adaptive token-level noise rescheduling (Ye et al., 2025). Efficient-DLM combines 500B tokens of continued training with blockwise attention and position-dependent masking (Fu et al., 2025), whereas iLLaDA returns to from-scratch diffusion pretraining at substantially larger scale (Nie et al., 2026). Diffusion conversion has also been demonstrated at larger sparse scales by RND1 (30B parameters), LLaDA2.0 (16B and 100B), and DiffusionGemma (25.2B, using uniform diffusion) (Chandrasegaran et al., 2025; Bie et al., 2025; DiffusionGemma Team, 2026).

Hybrid and blockwise models offer a different compromise between bidirectionality and deployability. Block Diffusion is autoregressive across blocks but diffusive within each block, enabling variable-length generation and inter-block KV caching (Arriola et al., 2025). Fast-dLLM v2 converts Qwen2.5 models using full-parameter fine-tuning with approximately 1B tokens and combines blockwise complementary masking with hierarchical caching (Wu et al., 2025). These systems retain more left-to-right structure than our fully bidirectional answer block, making them especially relevant to speed, caching, and arbitrary-length generation but not suited as direct controls for our intervention.

LAD first showed that LoRA can induce diffusion generation, using a hybrid objective that combines masking with token swaps, duplications, and span shifts (Kuiper et al., 2025; Hu et al., 2022). BYOD instead tests whether a simpler IID absorbing-mask objective can transfer reliably across model families. Other LoRA-based diffusion work addresses different goals: Uno learns diffusion-draft adapters while retaining the AR model as verifier, the dLLM framework uses LoRA to fine-tune already-diffusive checkpoints rather than for its Tiny-A2D conversion, and NaRA adapts existing diffusion models with noise-conditioned LoRA (Sahoo et al., 2026; Zhou et al., 2026; Wang et al., 2026b). Thus, their LoRA components augment already diffusively generating models, rather than converting AR model families into standalone diffusion models.

Table 1 places (estimates of) the training budgets of models of comparable size to our intended conversion targets side by side. For AR-to-diffusion conversion methods, these budgets exclude the substantial earlier cost of training the AR parent; for from-scratch methods they include diffusion pretraining. Dollar and operational-emissions columns are rental-equivalent estimates under one documented pricing and energy scenario, not the authors’ actual bills or measured footprints; Appendix A gives the complete calculation and its limitations.

Our study has explicitly chosen to hold the adaptation mechanism as constant as the architectures permit, repeat it across four families, and benchmark each converted model against the exact AR checkpoint from which it originated. This paired comparison is essential because it allows us to compare our results directly with the prior capability of the parent model.

Table 2: Diffusion model names matched to parent models, their trainable parameter counts, and measured time for 25,000 updates. All trainable parameters are rank-1024 Q/V LoRA weights.

Diffusion model	AR parent	Total parameters	Trainable	Trainable (%)	Time (h)
BYOD-Gemma	Gemma 2 9B IT	10.732B	572.5M	5.33	4.0
BYOD-Qwen	Qwen2.5 7B Instruct	7.939B	323.0M	4.07	3.1
BYOD-Ministral	Ministral 8B Instruct 2410	8.511B	490.7M	5.77	3.4
BYOD-Llama	Llama 3.1 8B Instruct	8.466B	436.2M	5.15	3.2

3 METHOD

3.1 MASKED-DIFFUSION TRAINING OBJECTIVE

The training objective teaches the adapted model to reconstruct masked response tokens while conditioning on an uncorrupted instruction.

Let p be the instruction, $x_0 = (x_0^1, \dots, x_0^L)$ the clean response region, A the answer positions, and P the EOS-padding positions. For each example, we sample $t \sim \mathcal{U}[10^{-3}, 1]$ and independently mask each position in $A \cup P$ with probability t , yielding the corrupted sequence x_t and masked set M_t . For either region $R \in \{A, P\}$, we calculate the same-position reconstruction loss

$$\ell_R(\theta) = -\frac{1}{t \max(1, |R|)} \sum_{i \in M_t \cap R} \log p_{\phi, \theta}(x_0^i | p, x_t). \quad (1)$$

We then combine the answer and padding losses:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, x_t} [0.9\ell_A(\theta) + 0.1\ell_P(\theta)]. \quad (2)$$

Here, $p_{\phi, \theta}$ is the conditional token distribution, ϕ denotes the frozen parent parameters, and θ the LoRA parameters. The first terminating EOS is included as a genuine answer token; all subsequent EOS tokens are treated as padding. The factor $1/t$ corrects for the sampled masking rate. Prompt and chat-template tokens remain visible to the model and are not part of the loss. Training masks are resampled online; validation masks are fixed by example index and seed.

3.2 AUTOREGRESSIVE-TO-DIFFUSION CONVERSION

To convert an autoregressive model to a diffusion-style model, we replace the causal mask in every layer with a bidirectional attention mask, so every prompt, answer, and EOS-padding position can attend to every position. We attach LoRA adapters to every self-attention query and value projection, with rank $r = 1024$, scaling $\alpha = 1024$, and dropout zero. The rest of the model architecture and output vocabulary remain unchanged. Backbone weights, token embeddings, the language-model head, biases, and normalization layers are frozen. The pretrained output head is also retained unchanged: whereas autoregressive training compares each position’s logits with the next token through shifted labels, BYOD removes this label shift and predicts the clean token at the same position from the bidirectionally attended, corrupted sequence. Each model reuses an existing single-token string from its tokenizer (MASK for Gemma, Llama, and Qwen; <?> for Ministral), avoiding vocabulary expansion.

We fix the LoRA rank across model families to retain one architecture-independent recipe. Matching trainable parameter counts would require model-specific ranks and would still not control for differences in depth, width, and attention design. This results in variation in trainable parameters, which is therefore a potential confounder in cross-family comparisons. Every conversion was trained on one NVIDIA RTX PRO 6000 GPU in Google Colab, without distributed parallelism; Table 2 reports parameter counts and conversion times.

For deployment, each adapter can be folded into its parent projection as $W_{\text{merged}} = W + (\alpha/r)BA$, where W is the frozen projection and A and B are the learned low-rank matrices. This removes the adapter branch while retaining the bidirectional attention intervention. Appendix B evaluates whether merging changes model behavior.

User: Replace the masked token in the partial assistant response with exactly the word written after "COPY:". Copy the spelling exactly. Do not use a synonym or a related word.

Initial masked model input:
 Answer: [MASK]. COPY: apple.

Expected model response:
 Answer: apple. COPY: apple.

Figure 2: Example input and expected reconstruction for the bidirectional attention test. Only the masked position is predicted.

3.3 TRAINING DATA

We construct a one-million-example mixture containing 45% general instruction following, 18% multiple-choice reasoning, 18% mathematics, and 19% code. Within these groups, the source weights are: Clean-Instruct-3M/Tulu 3/Alpaca-GPT4/Alpaca at 40/35/20/5 (Crumb, 2023; Lambert et al., 2024; Peng et al., 2023a; Taori et al., 2023); MMLU/HellaSwag/ARC-Easy at 73/25/2 (Hendrycks et al., 2021a; Zellers et al., 2019; Clark et al., 2018); Orca-Math/GSM8K at 95/5 (Mitra et al., 2024; Cobbe et al., 2021); and OpenCoder/MBPP at 99.7/0.3 (Huang et al., 2024; Austin et al., 2021b). When applicable, only the training splits are used. We deduplicate normalized prompts, exclude exact matches with held-out benchmark prompts, and retain only examples that fit the 256-token training length. Each model receives the same shuffled examples, retokenized with its native chat template and tokenizer. Prompt tokens remain clean; only answer and EOS-padding positions can be corrupted.

The four primary models see the same dataset prefix. BYOD-Llama-50k uses the same fixed shuffle for a longer, 800,000-example run with newly sampled corruptions. Exact run sizes and optimization settings are given in Appendix C.

3.4 INFERENCE

Inference begins from a fixed-length, fully masked answer block. At each denoising step, the model predicts all masked positions, retains the most confident predictions, and remasks lower-confidence positions; accepted tokens remain unmasked. This can resolve several output tokens per network evaluation. Task-specific lengths, denoising budgets, temperatures, and EOS/EOT handling are given in Appendix D.

4 EVALUATION

4.1 PAIRED MODELS AND BASELINES

We refer to the four conversions as BYOD-Gemma, BYOD-Qwen, BYOD-Ministral, and BYOD-Llama. For Llama, we distinguish the primary and longer-trained checkpoints as BYOD-Llama-25k and BYOD-Llama-50k. We compare each model with its unchanged AR parent. LLaDA 8B Instruct serves as an external diffusion baseline in the same harness; results reported by other papers are presented separately because their protocols differ. We use Llama for the longer run and checkpoint-wise analysis because DiffuLLaMA, LLaDA, and iLLaDA use Llama or closely related Llama-style architectures, making it the clearest basis for comparison with prominent general-purpose DLMs (Gong et al., 2025; Nie et al., 2025; 2026). We also compare BYOD-Llama-25k before and after folding its LoRA weights into the base model. The nine standard task benchmarks use greedy token selection and low-confidence remasking, while open-ended generation uses temperature 0.7 and delayed EOS/EOT acceptance. Complete settings are reported in Appendix D.

4.2 TASK BENCHMARKS

Our benchmark set contains the validation/test sets of MMLU, MMLU-Pro, HellaSwag, ARC-Challenge, GSM8K, MATH-500, GPQA-Main, HumanEval, and MBPP (Hendrycks et al., 2021a; Wang et al., 2024; Zellers et al., 2019; Clark et al., 2018; Cobbe et al., 2021; Hendrycks et al., 2021b; Rein et al., 2023; Chen et al., 2021; Austin et al., 2021b). We score multiple-choice tasks by option

Table 3: Exact-match/pass@1 accuracy in our matched harness. D denotes a converted diffusion model and AR its unchanged parent. The Llama D-25k and D-50k columns share the same AR parent. Among common-harness diffusion columns, bold marks the best result and underline the best BYOD result (ties retained); published LLaDA is protocol-separated and excluded from highlighting.

Task	Gemma		Qwen		Ministral		Llama			LLaDA	
	D	AR	D	AR	D	AR	D-25k	D-50k	AR	Ours	Published
<i>General knowledge and reasoning (multiple choice)</i>											
ARC-C	.864	.888	.840	.920	.832	.740	.832	.792	.792	.864	.885
GPQA	<u>.320</u>	.280	.288	.400	.272	.340	.296	.272	.320	.312	.333
HellaSwag	.752	.696	.728	.780	.768	.700	.704	.688	.608	.696	.746
MMLU	.688	.648	.568	.660	<u>.576</u>	.520	.592	.608	.536	.560	.655
MMLU-Pro	<u>.368</u>	.448	.280	.400	.360	.320	.304	.304	.416	.424	.370
<i>Math (open ended)</i>											
GSM8K	.640	.872	.608	.900	.656	.860	.656	.560	.864	.648	.694
MATH-500	.296	.600	.352	.780	.208	.520	.256	.216	.480	.344	.319
<i>Code generation (open ended)</i>											
HumanEval	.380	.740	.296	.960	.500	.940	.312	.336	.560	.400	.494
MBPP	.460	.740	.440	.820	.520	.720	.432	.512	.712	.424	.410
Average	.530	.657	.490	.736	.521	.629	.487	.476	.588	.519	.545

label, GSM8K by its final number, MATH by its normalized final boxed expression, and code tasks by whether the generated program passes the supplied tests in an isolated subprocess.

4.3 OPEN-ENDED GENERATION

We assess open-ended generation with perplexity and Distinct-1/2/3. Standard masked diffusion does not directly provide the left-to-right sequence likelihood used for AR perplexity, and its usual evidence lower bound is not a proper test-time likelihood (Turok et al., 2026). We therefore follow prior DLM work and use frozen Microsoft Phi-4 14B as an external perplexity scorer (Gong et al., 2025). We pool negative log-likelihoods across all non-padding tokens of all generated answers before exponentiating, so every token has equal weight. Because this score does not measure diversity, we also report sliding model-token Distinct-1/2/3 percentages after removing special tokens. Higher Distinct- n indicates less local repetition (Li et al., 2016); in our experiments, it exposed cases where perplexity improved because output became more repetitive rather than better.

4.4 BIDIRECTIONAL ATTENTION TEST

We test whether the converted diffusion models allow bidirectional information flow with a simple right-context copy task (Appendix F). Most of the response is pre-filled, and the model must fill an earlier masked position by copying a word shown later in the response. A causal model cannot change its prediction at an earlier position based on a later token, so sensitivity to this clue directly tests functional bidirectional attention. Figure 2 shows one test item and its expected reconstruction.

5 RESULTS

5.1 BENCHMARK ACCURACY AND BASE-MODEL DEPENDENCE

Table 3 shows that BYOD-Gemma has the highest diffusion model average, also exceeding LLaDA in the matched harness. The AR parents perform better on average, but the conversion gap varies by task and backbone. BYOD-Llama-50k benchmark results remain largely similar to BYOD-Llama-25k, so additional training does not improve factual accuracy uniformly.

Figure 1 shows that the AR advantage is largest in mathematics and code, while the relative strengths of the converted backbones are task dependent. LLaDA’s score in our harness is 2.6 points below its published average, underscoring the importance of prompts, subsets, decoding, and answer extraction.

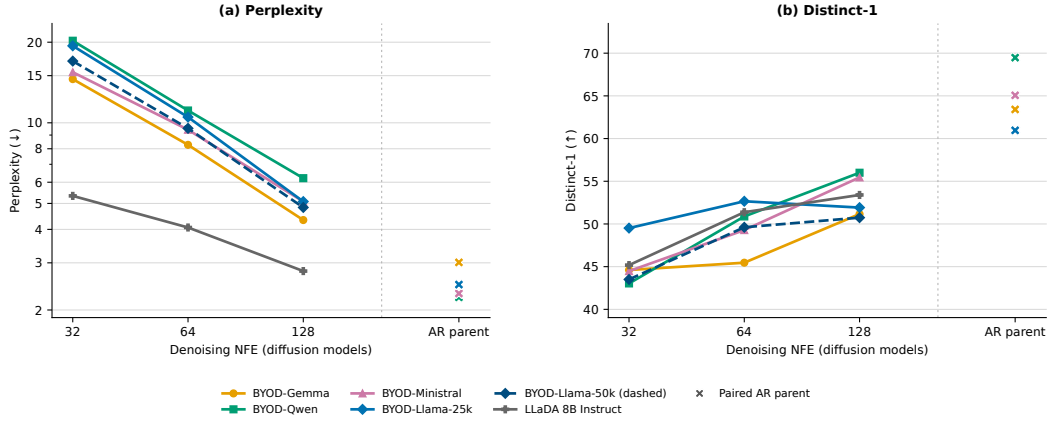


Figure 3: Visualization of the open-generation results in Appendix Table 10. Lines connect each diffusion model’s results at 32, 64, and 128 NFE; crosses to the right show the corresponding AR-parent reference without treating AR decoding as an NFE condition. The dashed dark-blue curve is BYOD-Llama-50k. BYOD-Llama-25k and BYOD-Llama-50k share one AR parent, and LLaDA has no paired AR reference. Perplexity is shown on a logarithmic axis.

We therefore treat the matched-harness result as primary and report further comparison to other external diffusion models separately in Appendix G. LLaDA is our only directly tested external DLM because its masking objective and benchmark coverage provide the cleanest comparison. Finally, the merged BYOD-Llama model remains broadly similar across factual and open-generation measures; Appendix B reports the full ablation.

5.2 OPEN-ENDED GENERATION WITH REDUCED DENOISING STEPS

The converted models have 2.7–5.0 times the perplexity of their AR parents for different NFE budgets, and generally lower Distinct- n scores (Appendix Table 10). Figure 3 shows that perplexity and Distinct-1 generally improve with additional NFE. The paired AR parents retain both lower perplexity and higher diversity. LLaDA also has lower perplexity than every BYOD model. Appendix Table 10 reports exact perplexity and Distinct-1/2/3 at 32, 64, and 128 NFE.

5.3 FURTHER TRAINING

Figure 4 shows that validation loss decreases steadily. However, open-ended fluency metrics do not move monotonically, although a general improvement trend is visible over the course of training. Validation loss alone might therefore not be a reliable checkpoint-selection metric for open generation, especially at low NFE.

5.4 BIDIRECTIONAL ATTENTION TEST

The bidirectional attention test showed that all models use right-context information. Adding the later clue changes every model’s greedy prediction on every test item, whereas no model copies the target after the clue is removed (Appendix F). Gemma, Ministral, and LLaDA copy the clue perfectly; Llama is less consistent and Qwen is substantially worse.

6 DISCUSSION

The central result is that high-rank attention-only LoRA can convert four causal instruction models into bidirectional masked generators in a few hours on one GPU. Their benchmark average is competitive with LLaDA in our matched harness, and the right-context test confirms functional bidirectional conditioning. Conversion quality nevertheless depends on the backbone: Gemma

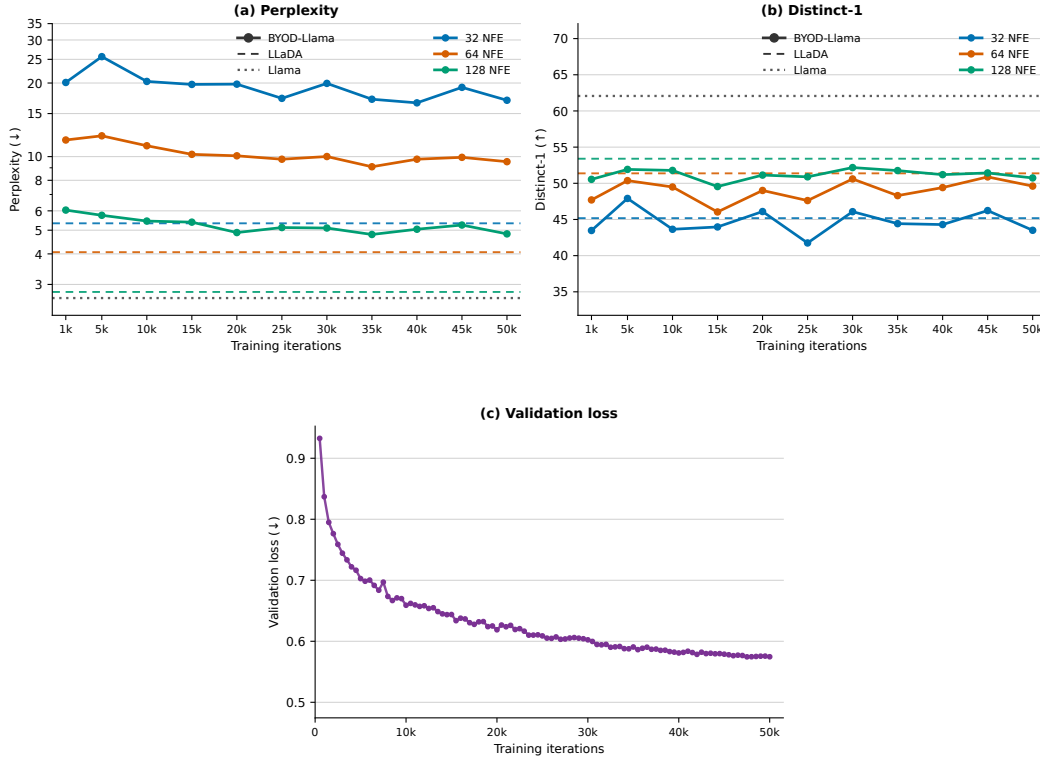


Figure 4: BYOD-Llama-50k training diagnostics. (a) Token-weighted Phi-4 perplexity for 128-token generations, shown on a logarithmic y-axis. (b) Mean sliding model-token Distinct-1 for the same generations, reported as a percentage. Color-matched dashed lines are fixed LLaDA references; the dotted gray line is the unchanged Llama AR parent. (c) Validation loss.

performs best, while Qwen’s stronger AR parent does not produce the strongest conversion. Parent quality, architecture, model size, and absolute adapter size remain confounded.

The single-Colab workflow broadens access to DLM experimentation, although it is not cost-free. The public demos also make grammaticality, coherence, and denoising failures easier to inspect than aggregate metrics alone; representative trajectories appear in Appendix H. Their EOS-aware early stopping can finish before the fixed denoising budget when the pre-EOS answer stabilizes, following related work on variable-length and token-level stopping (Yang et al., 2025; Liu et al., 2026; Wang et al., 2026a; Kohut et al., 2026).

The high LoRA rank is larger than a conventional task adapter but reflects the harder objective of learning a new generation process. LAD and our preliminary experiments found rank 1024 preferable to smaller ranks, without consistent gains at 2048 (Kuiper et al., 2025). This overhead is temporary: merging returns Llama from 8.466B to 8.030B parameters while leaving measured behavior broadly stable (Appendix B).

The main remaining weakness is generation quality. Perplexity and Distinct-1/2/3 trail both the AR parents and LLaDA, especially when NFE is below output length. Moreover, validation loss improves more steadily than generation metrics. Longer training alone may therefore be insufficient; useful ablations include larger effective batches, alternative denoising targets, EOS-padding supervision, and adapters in key/output projections, MLPs, or normalization layers.

Perplexity is not sufficient for model selection. An external AR scorer can favor repetitive high-probability text, while Distinct- n captures repetition but not factuality, grammar, or coherence (Gong et al., 2025; Turok et al., 2026). We therefore combine perplexity, Distinct-1/2/3, task accuracy, and generated samples rather than treating any one metric as definitive.

BYOD trades maximum quality for a much smaller conversion budget. Full-model systems use far more data and often add architecture-specific training or decoding machinery, so their published scores are not controlled comparisons. Our direct LLaDA run is therefore the primary external baseline; the remaining literature values in Appendix G provide context only. Table 1 shows the scale difference, but its cost and emissions estimates do not replace measured power or location-specific carbon accounting.

7 LIMITATIONS

The main limitation is that efficient conversion does not yet produce generation quality on par with the AR parents or LLaDA, especially when NFE is below output length. We also measure inference budget in network evaluations rather than wall-clock latency. Fewer sequential evaluations do not guarantee a speedup because each diffusion step processes the full answer block. Our efficiency claim therefore concerns training the conversion, not measured inference acceleration.

The cross-backbone comparison is not a controlled architecture study. The parents differ in size, tokenizer, pretraining, alignment, and attention design. Fixing LoRA rank also gives each backbone a different absolute adapter budget. Gemma’s leading result therefore cannot be attributed separately to architecture, parent quality, model size, or trainable parameter count.

Finally, the automatic generation metrics cover only part of text quality. Phi-4 perplexity reflects the preferences of an external AR model, while Distinct- n detects local repetition but not factuality, grammar, or coherence. Published results from other DLMs use different protocols; only LLaDA was rerun in our harness. We also did not measure energy use or carbon intensity directly, so the resource analysis supports relative scale and accessibility claims rather than a measured environmental footprint.

8 CONCLUSION

BYOD provides a common, efficient route from instruction-tuned AR checkpoints to masked diffusion generators across Gemma, Llama, Qwen, and Ministral. The experiments establish that diffusive behavior, functional bidirectional conditioning, and substantial parent-model knowledge can be obtained in a few hours on one Google Colab GPU. They also expose the remaining challenge: fluency and diversity lag behind AR parents and LLaDA, especially when NFE is below output length. The next step is therefore not merely more compute, but determining whether longer training, larger effective batches, alternative objectives, or a broader adapter placement can close this gap without sacrificing BYOD’s accessibility. In its present form, BYOD makes that experimentation feasible for researchers who cannot retrain a foundation model or provision a large accelerator cluster.

REPRODUCIBILITY STATEMENT

Training and evaluation code is available at <https://github.com/RuurdKuiper/lad-generics>. Public model cards and interactive demos for all four conversions are available under <https://huggingface.co/Ruurd>; each model card links to its corresponding full-precision ZeroGPU demo. The main text specifies the objective, trainable modules, data volume, optimization, generation, benchmarks, and scoring. The accompanying artifacts include resolved configurations, parameter audits, checkpoint identifiers, loss logs, generated samples, and benchmark manifests.

ETHICS STATEMENT

This study uses existing language models and public benchmark or instruction datasets and does not involve human subjects. Converted models inherit factual errors, biases, unsafe behaviors, licensing constraints, and memorization risks from their parent checkpoints and training data. We report aggregate task performance rather than claiming safety or deployment readiness; release artifacts should preserve upstream licenses and document the data provenance and intended research use.

AI USE STATEMENT

Generative AI tools were used to assist software development and debugging, design and review of experiments, creation of analysis scripts, interpretation of results, literature discovery and summarization, and drafting and editing this manuscript. The authors manually reviewed AI-assisted code and prose, checked numerical claims against saved experimental artifacts, and verified literature claims and citations against primary sources. No benchmark value was entered solely from an unsupported model response. The authors take responsibility for the final content, claims, code, and artifacts.

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A COST AND OPERATIONAL-EMISSIONS ESTIMATION

Table 1 compares the cost of training a model that produces diffusion behavior. It should thus be taken into account that some of these models build on an AR parent, whereas others are trained from scratch. It also excludes exploratory runs, failed jobs, dataset construction, evaluation, storage, and inference. The estimates must therefore not be interpreted as total project or lifecycle footprints.

For BYOD we use the observed 3.2 GPU-hours for the conversion of a Llama 3.1 8B model for diffusive generation. For a prior work that reports GPU-hours, we retain that value. Where a paper reports model size but not time, we approximate training compute as $C \approx 6ND$ FLOPs (Hoffmann et al., 2022), where C is total training compute, N is the number of model parameters, and D is the number of processed training-token positions. We then calculate

$$G_{\text{est}} = \frac{6ND}{0.40(989 \times 10^{12})3600}, \quad (3)$$

where G_{est} is aggregate estimated GPU-hours, 0.40 is the assumed model FLOPs utilization (MFU), 989×10^{12} is the NVIDIA H100’s dense BF16 Tensor Core throughput in FLOP/s (NVIDIA, 2023), and 3600 converts seconds to hours. The 40% MFU assumption is consistent with the 39.0–45.0% BF16 MFUs reported by FP8-LM for GPT-style 7B, 13B, and 175B training on H100 GPUs (Peng et al., 2023b). Actual utilization varies considerably with architecture, sequence length, parallelization, communication, activation recomputation, and implementation efficiency. The assumption is therefore a standardized order-of-magnitude convention rather than a reconstruction of any authors’ cluster. It yields the daggered values in Table 1. The Dream and iLLaDA estimates cover their main 580B- and 12T-token stages but omit the smaller instruction-tuning additions, so they are lower bounds.

Table 4 calibrates the same calculation against studies that report both a usable token count and elapsed accelerator-hours (Nie et al., 2025; Wu et al., 2025). For LLaDA, it predicts approximately

77,519 GPU-hours, compared with 130,000 reported H800 GPU-hours, corresponding to an implied utilization of 23.9%. Thus 40% can be optimistic about achieved utilization and correspondingly conservative with respect to estimated accelerator-hours, rental cost, and emissions. Fast-dLLM v2 displays a still larger gap. Its specialized training concatenates clean and noised views and uses a nonstandard attention pattern, illustrating that an “implied MFU” from *6ND* is a calibration diagnostic, not a measured hardware-utilization statistic.

Table 4: Calibration of the *6ND* estimator against reported accelerator-hours. Predictions use 40% MFU and the stated accelerator’s dense-BF16 peak (989 TFLOP/s H100-class proxy for H800; 312 TFLOP/s for A100) (NVIDIA, 2023; 2021). “Implied MFU” is the utilization that would reconcile *6ND* with reported time; it is not a measured MFU.

System	Parameters	Tokens	Pred. GPU-h	Reported GPU-h	Implied MFU
LLaDA	8B	2.30T	77,519	130,000	23.9%
Fast-dLLM v2	7B	1.31B	123	768	6.4%

We convert BYOD time to a rental-equivalent cost using the September 2026 AWS US-East on-demand price of \$3.36 per hour for a one-GPU `g7e.2xlarge`, which contains the same 96-GB NVIDIA RTX PRO Server 6000 accelerator used in our Colab run (Amazon Web Services, 2026b). For other systems we use one consistent current major-cloud proxy: \$5.191 per accelerator-hour for an AWS `p5.4xlarge` H100 Capacity Block in US East (Amazon Web Services, 2026a). Thus these figures answer “what would the stated or estimated accelerator-hours cost at a public contemporary rate?”, not what the original authors paid. They exclude storage, egress, idle reservation time, taxes, and discounts.

Operational emissions are estimated as energy use multiplied by the data center’s power usage effectiveness (PUE) and the grid carbon intensity (Lannelongue et al., 2021; Patterson et al., 2021):

$$\hat{E}_{\text{CO}_2\text{e}} = G P_{\text{GPU}} \text{PUE} I_{\text{grid}}. \quad (4)$$

Here $\hat{E}_{\text{CO}_2\text{e}}$ is estimated operational emissions in kg CO₂e, G is aggregate GPU-hours, P_{GPU} is assumed per-GPU power draw in kW, PUE is the dimensionless data-center power usage effectiveness, and I_{grid} is grid carbon intensity in kg CO₂e/kWh. We use 0.60 kW for RTX PRO 6000 runs, matching NVIDIA’s stated maximum board power (NVIDIA, 2025), and a 0.70-kW H100-class proxy for all other entries except the reported A100 run, for which we use 0.40 kW (NVIDIA, 2023; 2021). We assume PUE 1.09, Google’s reported fleet-wide value (Google Cloud, 2025), and the IEA’s 2024 global-average generation intensity of 0.445 kg CO₂e/kWh (International Energy Agency, 2025). For example, one BYOD stage gives $3.2 \times 0.60 \times 1.09 \times 0.445 = 0.93$ kg CO₂e. These are location-independent GPU-only operating scenarios: they assume full rated GPU power, omit host CPU/memory/network energy and embodied hardware emissions, and do not account for renewable-energy procurement. Direct power telemetry and the datacenter’s time- and location-specific marginal carbon intensity would be required for a measured footprint.

B ADAPTER-MERGING ABLATION

High-rank LoRA increases the parameter count only while the adapter remains separate. To test whether removing this overhead changes behavior, we evaluate the same BYOD-Llama-25k checkpoint with the adapter loaded and after its weights are folded into the base projections. Tables 5 and 6 report open-generation and factual results, respectively.

Table 5: Open-generation comparison of the adapter-loaded and merged Llama checkpoints. PPL is token-weighted; Distinct- n is averaged over answers and reported as a percentage. Both use 128 generated tokens; D-1/2/3 use 32 NFE.

Representation	Parameters	PPL ₃₂ ↓	PPL ₁₂₈ ↓	D-1%↑	D-2%↑	D-3%↑
Adapter loaded	8.466B	19.381	3.753	49.5	78.8	89.9
Merged standalone	8.030B	19.283	3.720	49.8	80.7	91.8

Table 6: Factual benchmark accuracy (%) before and after merging in the matched harness; the mean is unweighted across the nine tasks.

Task	Adapter	Merged	Task	Adapter	Merged
ARC-C	82.8	82.0	MATH	22.0	22.0
GPQA	28.0	30.0	MBPP	46.0	44.0
GSM8K	65.2	58.0	MMLU	61.2	64.0
HellaSwag	73.2	68.0	MMLU-Pro	33.2	28.0
HumanEval	50.0	48.0	Mean	51.3	49.3

Merging is an algebraic reparameterization rather than an additional training stage: each base projection W is replaced by $W + (\alpha/r)BA$, where A and B are the trained rank- r LoRA matrices and α is their scaling parameter. Subject to floating-point rounding, evaluating the folded projection is equivalent to evaluating the base and LoRA branches separately. The open-generation metrics are slightly better after merging, whereas the factual benchmarks show mixed task differences and a 2.0-point lower macro mean. These reruns therefore support preservation of overall behavior, but are not precise enough to attribute task-level changes to merging; a logit-level deterministic parity test would isolate numerical merge error from generation and benchmark variance.

C COMPLETE TRAINING CONFIGURATION

Table 7 records the common resolved configuration of the four primary conversions, including optimization, precision, checkpointing, and validation. We use AdamW with BF16 master weights and Transformer Engine HYBRID FP8 execution.

BYOD-Llama-50k keeps the same batch size and objective but processes 800,000 examples over 50,000 updates, validates every 500 updates, and uses a linear learning-rate decay from 10^{-5} to 10^{-6} . This run supplies the BYOD-Llama-50k results and the diagnostics in Figure 4.

D COMPLETE INFERENCE CONFIGURATION

Table 8 separates task-specific diffusion budgets from the training configuration and gives the evaluation sample count in one place. Unless overridden in the table, diffusion evaluation uses the official LLaDA-style fixed-transfer sampler, greedy predictions (temperature 0), low-confidence remasking, classifier-free guidance scale 0, one full-answer block, and no repetition penalty. Delayed EOS/EOT means that a predicted terminator is assigned the lowest retention confidence and may therefore be predicted but is accepted late; HumanEval instead suppresses terminator logits. The open-ended diagnostic uses temperature 0.7. Archived run manifests take precedence over mutable notebook defaults.

Autoregressive parents are decoded greedily. Their maximum-new-token budgets are reported separately because an AR token budget is not an NFE budget and should not be read as computational equivalence.

E COMPLETE OPEN-GENERATION RESULTS

Table 7: Full common training configuration. Model-specific parent checkpoints, mask strings, and parameter counts appear in Table 2.

Setting	Value	Setting	Value
Training examples	400,000	Optimizer updates	25,000
Maximum sequence length	256	Dynamic padding multiple	16
Batch size	16/16	Gradient accumulation	1
Optimizer	AdamW	Peak learning rate	10^{-5}
Schedule	cosine	Warmup	100 updates
Weight decay	0.01	Maximum gradient norm	0.5
Training precision	FP8 HYBRID	Master/evaluation precision	BF16
Gradient checkpointing	disabled	Selected-logit optimization	enabled
Corruption	IID mask-only	Noise time	$t \sim \mathcal{U}[0.001, 1]$
Mask-eligible region	answer + EOS padding	Loss region	corrupted answer + EOS padding
Answer/padding loss weights	0.9/0.1	Genuine answer EOS	included
LoRA targets	query/value projections	LoRA rank/scaling/dropout	1024/1024/0
Trainable normalization	disabled	Quantization	none
Validation	200 examples/250 updates	Checkpoint interval	1,000 updates
Data-loader workers	4	Prefetch factor	4

Table 8: Task-specific masked-diffusion inference settings in the validation notebook. NFE is the number of denoising network evaluations.

Task	Output tokens	NFE	Block length	Temperature	Terminator handling	Samples
Open-ended	128	128/64/32	128	0.7	delayed retention	125
MMLU	64	64	64	0	ordinary	125
MMLU-Pro	128	128	128	0	delayed retention	125
HellaSwag	64	64	64	0	ordinary	125
ARC-Challenge	64	64	64	0	ordinary	125
GSM8K	256	256	256	0	delayed retention	125
MATH-500	256	256	256	0	delayed retention	125
GPQA-Main	128	128	128	0	delayed retention	125
HumanEval	256	256	256	0	delayed retention	125
MBPP	128	128	128	0	delayed retention	125
Bidirectional attention	1 masked word	1	–	0	–	20

Table 9: Maximum new tokens for paired greedy AR evaluation.

Task	Tokens	Task	Tokens
Open-ended	128	MMLU	64
MMLU-Pro	2,048	HellaSwag	64
ARC-Challenge	64	GSM8K	256
MATH-500	256	GPQA-Main	128
HumanEval	256	MBPP	128

Table 10: Open-ended generation on a 128-token canvas. PPL_k is token-weighted Phi-4 perplexity and D-1_k is mean per-answer sliding model-token Distinct-1 at k NFE. Lower PPL and higher D-1 are better. AR parents generate autoregressively and therefore have no diffusion-NFE setting.

Model/checkpoint	Diffusion						AR parent	
	$\text{PPL}_{32} \downarrow$	$\text{D-1}_{32} \% \uparrow$	$\text{PPL}_{64} \downarrow$	$\text{D-1}_{64} \% \uparrow$	$\text{PPL}_{128} \downarrow$	$\text{D-1}_{128} \% \uparrow$	$\text{PPL} \downarrow$	$\text{D-1} \% \uparrow$
BYOD-Gemma	14.56	44.6	8.27	45.5	4.33	51.2	3.01	63.4
BYOD-Qwen	20.25	43.0	11.12	50.9	6.21	56.0	2.23	69.5
BYOD-Ministral	15.45	44.5	9.42	49.3	5.08	55.5	2.31	65.1
BYOD-Llama-25k	19.38	49.5	10.50	52.7	5.08	51.9	2.49	61.0
BYOD-Llama-50k	17.00	43.5	9.53	49.6	4.83	50.7	2.49	61.0
LLaDA 8B Instruct	5.33	45.2	4.07	51.4	2.79	53.4	–	–

F BIDIRECTIONAL ATTENTION TEST

This deterministic test uses ordinary single-token target words. Each partial response has the form “Answer:[MASK]. COPY: apple.”, with a different target in each example. The target is absent from the instruction and left context and appears explicitly after the mask. The converted model predicts the masked position by greedy argmax in one forward pass; no iterative sampling or generation settings are involved.

We score the same masked position twice: once with the later copy clue and once after deleting that clue. We report exact match for both conditions, their difference, and whether adding the clue changes the greedy prediction at all. Under causal attention, changing or deleting tokens to the right cannot affect logits at an earlier position, so both the exact-match gain and prediction-change rate must be zero. The latter separates use of right context from the stricter ability to follow the exact-copy instruction. This is a functional attention check, not a benchmark of reasoning or general infilling quality.

Table 11: Bidirectional attention test results. “With clue” predicts the earlier mask while the word is visible later; “without clue” removes that later word. Gain is the exact-match difference.

Model	With clue EM	Without clue EM	EM gain	Prediction changed
BYOD-Gemma	100%	0%	+100 pp	100%
BYOD-Qwen	20%	0%	+20 pp	100%
BYOD-Ministral	100%	0%	+100 pp	100%
BYOD-Llama-25k	90%	0%	+90 pp	100%
BYOD-Llama-50k	70%	0%	+70 pp	100%
LLaDA 8B Instruct	100%	0%	+100 pp	100%

G PROTOCOL-SEPARATED LITERATURE RESULTS

Table 12 places our primary Llama conversion beside values reported by other diffusion-model papers (Nie et al., 2025; Ye et al., 2025; Nie et al., 2026; Fu et al., 2025). The BYOD column uses our common harness, whereas every other column uses its source paper’s evaluation protocol. Differences in prompts, subsets, generation budgets, answer extraction, and execution environments prevent direct ranking; the table is included only to provide scale and task coverage. NR means that the source did not report a value for that task.

Table 12: Our Llama result and protocol-separated published diffusion-model accuracies. Bold marks the largest reported value in each row; it does not imply a controlled comparison. No cross-task averages are shown because task coverage differs by model.

Task	BYOD-Llama (ours)	LLaDA	Dream-Inst.	iLLaDA-Inst.	Efficient-DLM 8B
ARC-C	.832	.885	NR	NR	.619
GPQA	.296	.333	.330	NR	NR
GSM8K	.656	.694	.810	.890	.883
HellaSwag	.704	.746	NR	NR	.725
HumanEval	.312	.494	.555	.659	.683
MATH	.256	.319	.392	.567	.501
MBPP	.432	.410	.588	.580	.610
MMLU	.592	.655	.670	.716	.772
MMLU-Pro	.304	.370	.433	.523	NR

H INTERACTIVE DENOISING TRAJECTORIES

Figures 5 and 6 illustrate how the public BYOD-Gemma demo exposes the parallel refinement process under random and confidence-guided remasking. All panels use the displayed Amsterdam

prompt, the helpful-assistant system prompt, temperature 0.7, top- k 3, initial noise fraction 1.0, delayed EOS/EOT retention, permanent locked retention, and no visualization delay. Early stopping is disabled and block length equals the new-token canvas, ensuring that every panel displays its complete denoising budget. The demo randomizes the hidden sampling seed for every request; we resample the 64-NFE conditions to show readable positive examples, so these trajectories illustrate behavior rather than constitute a controlled quality comparison.

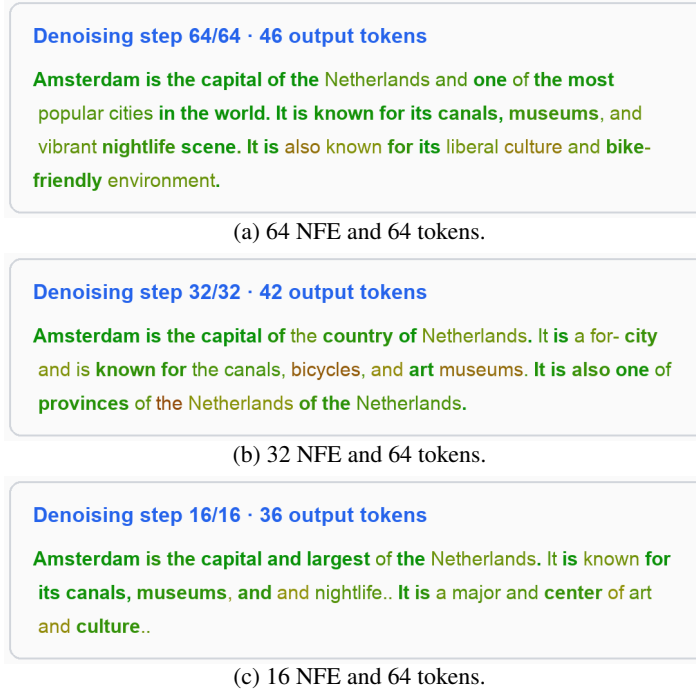


Figure 5: Confidence-guided BYOD-Gemma-2-9B trajectories at decreasing inference budgets on a 64-token canvas. Red-to-green coloring records each retained token’s sampling probability.

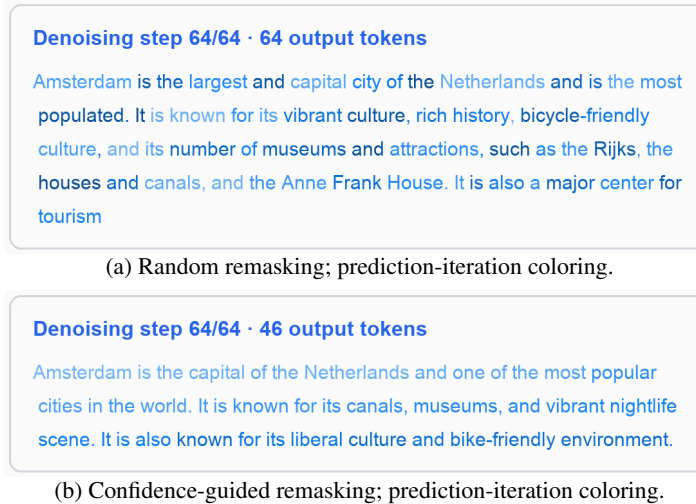


Figure 6: Remasking comparison for BYOD-Gemma-2-9B at 64 NFE on a 64-token canvas. Blue intensity records the iteration at which each retained token was predicted.